

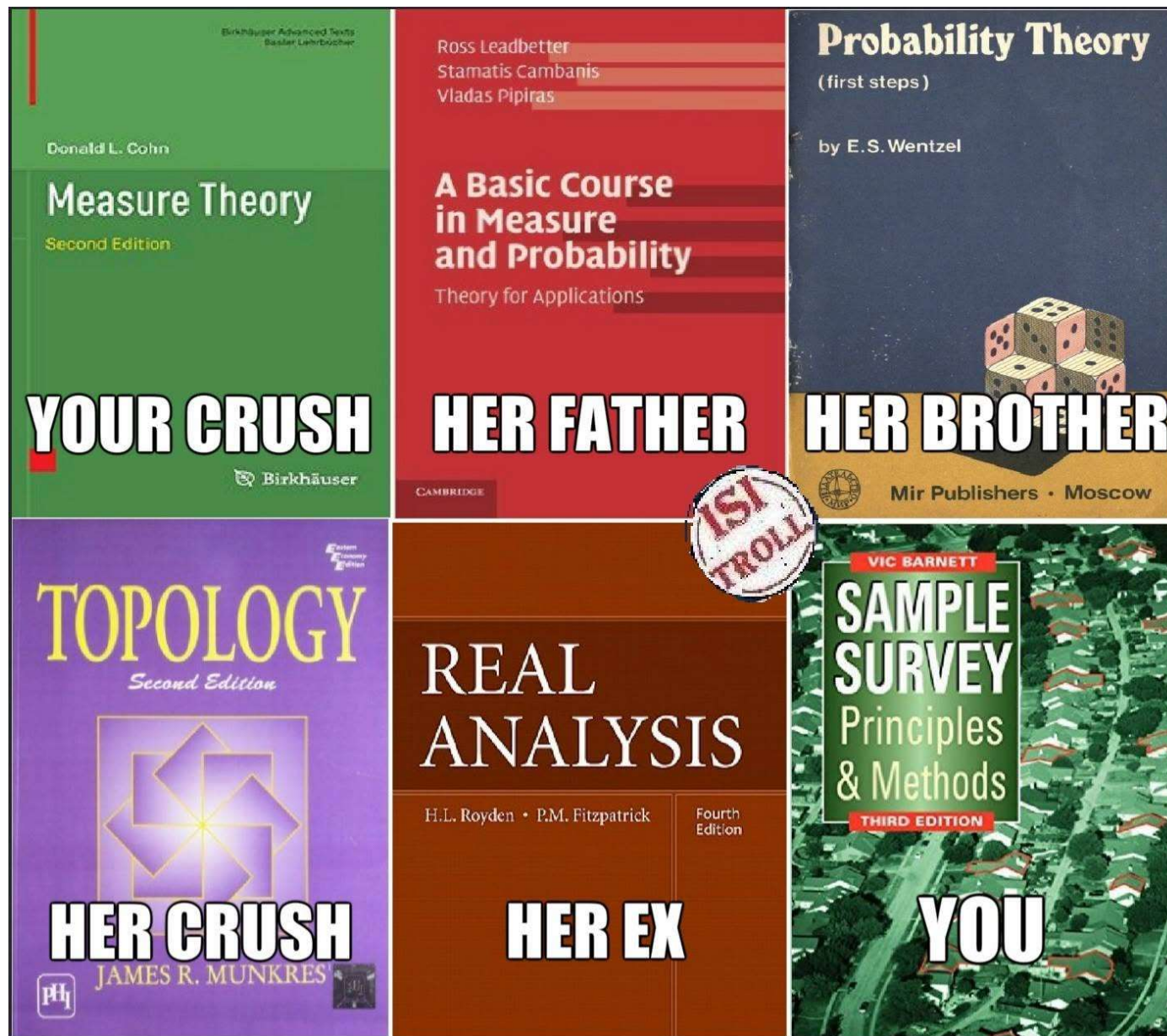
# Sample Surveys

## Class 1

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Fall 2026



An excerpt from *Introduction to Statistics* by Giri and Banerjee:

### § 2.1 DEFINITION OF STATISTICS

The word *statistics* is associated with the term *state* (indicating quantitative information relating to state administration). The term *statistics* was coined by the German scholar Gottfried Achenwall around the middle of the eighteenth century. In modern age, this word includes all branches of human activity concerned with numerical data.

*Statistics* is used as a plural noun and also as a singular noun.

In plural sense, numerical data, i.e., numerical statements of facts in connection with any field of human experience are called *statistics*. For example, production statistics of a factory over different years, statistics of birth and death of a country, etc.

The term *statistics*, in singular sense, means the aggregate of scientific methods which deals with the **collection**, presentation, analysis and interpretation of numerical data. This singular sense of *statistics* is used in the statement, “R. A. Fisher had contributed a lot in various branches of statistics”.

- Design of Experiments & Sample Surveys:
  - Deals with the **collection** of data, and more...

# Sample Surveys: ISI's DNA



- Mahalanobis's work in large-scale sample surveys in the 1930s and 40s put ISI Kolkata on the global map.
- Estimating Bengal's massive [jute production](#).
  - Sample survey was cheaper, faster, and more accurate than full government census.
- Birth of [National Sample Survey](#) (NSS, 1950).
  - The most comprehensive, multi-subject, continuous household survey in the world.
  - Template for the United Nation's global sampling guidelines.
- Theory and methods born from [practice](#).
  - Rigorous constrained optimization problems.
  - Interpenetrating Network of Sub-sampling (IPNS): Doing resampling before it was cool.
  - Pilot surveys: Doing sequential learning before it was cool.

# Questions



- How much does a typical family in a district spend each month on food, rent, and basic needs?
- What fraction of young adults in a state are currently working or looking for work?
- What is the average birth weight of newborns delivered in public hospitals across a state?
- What portion of a city's population developed antibodies after a virus outbreak?
- How many tigers or elephants are currently living in a national park reserve?
- What fraction of micro-transactions on a UPI app fail or run into errors on a given day?
- What is the average distance a student in a rural area has to walk to reach the nearest high school?
- Who do voters plan to support in an upcoming election?

# Marks distribution

- Midsem: 30%.
- Endsem: 50%.
- Final presentations: 20%.
  - Students divided into groups.
  - Topics not directly covered in class.
- Google classroom?

# The ABCD of Sample Surveys

## Example

The state health ministry is planning to launch a new health insurance scheme. To budget for this, they must estimate the **average monthly out-of-pocket medical expenditure** for **all adults in West Bengal**.

- **Observation unit:** An object on which a measurement is taken, e.g., an adult person.
  - Could be anything: an animal, a region, a hospital, a plot of agricultural land, a financial record.
- **Target population:** The ideal, complete collection of observation units (in a particular time period) we want to infer about, e.g., all adults in West Bengal.

## Survey by *complete enumeration* or *census*: study everybody

- Collect data on one or more variables of interest (e.g., average monthly out-of-pocket medical expenditure) for **all the units** in the population.
- Quite **hard**, **expensive**, and **time consuming** if the population is big.

Alternative option: Survey by *sampling*

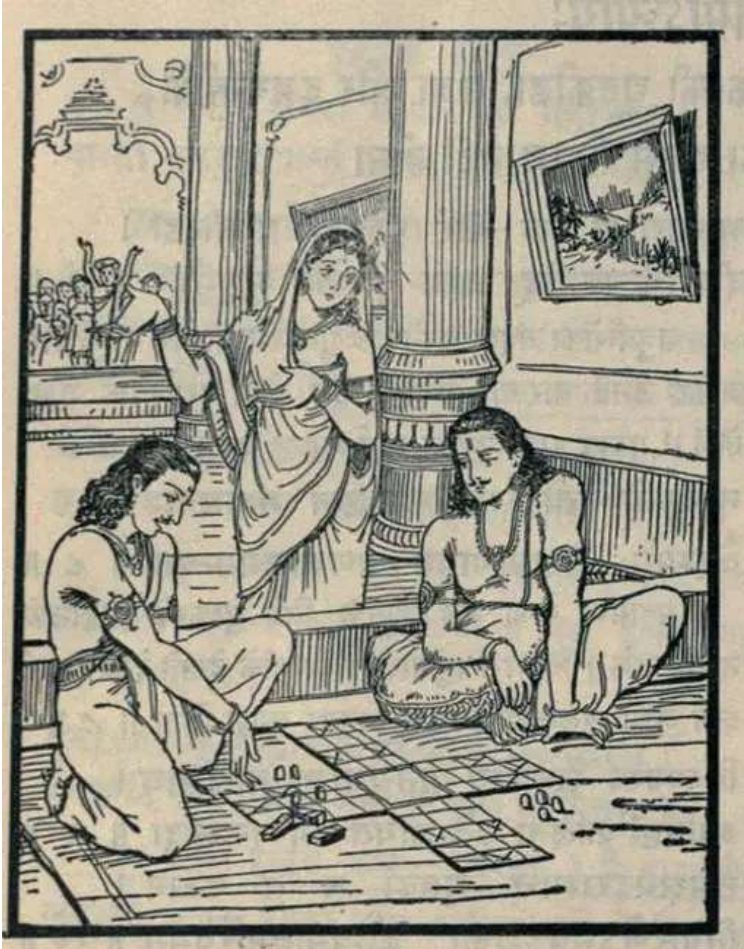
# The ABCD of Sample Surveys

- **Sample:** A subset of a population.
- **Sampled population:** The collection of all possible observation units that might have been chosen in a sample, i.e., the part of the target population that is accessible.
  - In an ideal survey, sampled population = target population.
  - In our example, it could be those adults living in registered households.
- **Sampling unit:** A unit that can be selected for a sample.
  - If we don't have the list of adults, households could serve as the sampling units, and the observation units are the adults living in the households.
- **Sampling frame:** The tangible list, map, database, or other specification of sampling units from which a sample may be selected.
  - E.g., the most recent West Bengal Electoral Roll.
  - Must have unique IDs or serial numbers.
  - Proper care should be taken to create the sampling frame.

## Sample survey

The process of selecting a sample, and collecting data on one or more variables of interest for these selected units, to estimate characteristics of the target population.

# Earliest instance of sampling

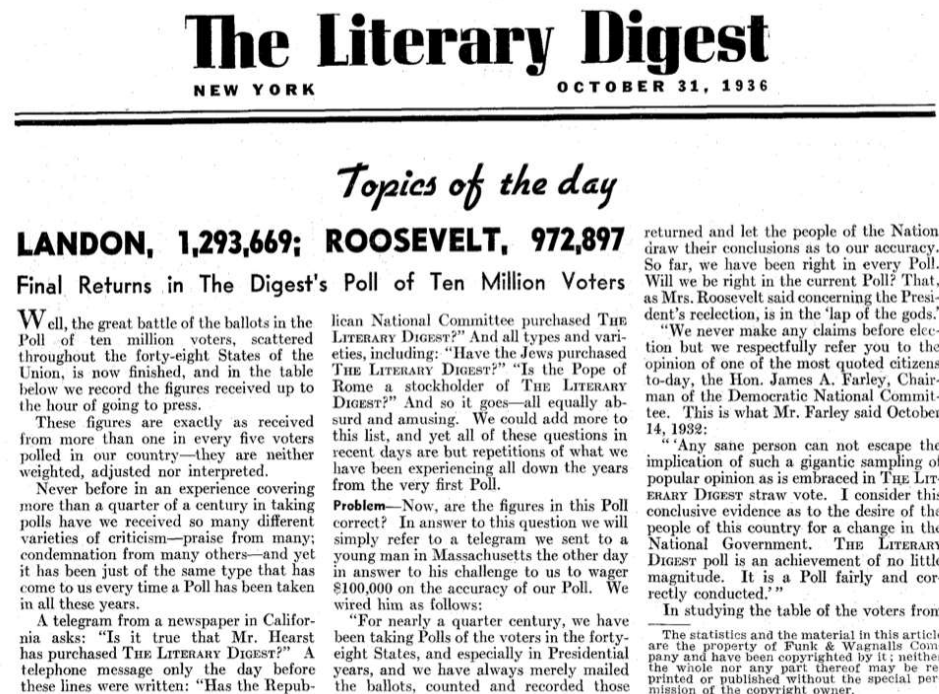


- *Mahabharata*: Story of King Nala and Rupa.
- Nala loses his kingdom in gambling, wanders in the forest, takes a job as a charioteer to Ayodhya king Rupa.
- On a journey, they pass a sprawling tree. By glancing at the tree, he tells Nala the total number of fruits and leaves on the entire tree.
- Nala chops down the tree and spends the whole night counting every single leaf and fruit. He gets amazed by the accuracy of Rupa's guess.
- Rupa reveals his secret: he gives his estimate based on examining one small branch.

*"Know me to be a knower of the secret of the dice and therefore adept in the art of enumeration."*

# Bad sample: The *Literary Digest* case

- One of the **worst** political predictions in history.
- Who will win the 1936 US presidential election?
  - Franklin Roosevelt (**democratic**) vs Alfred Landon (**republican**)
  - 10 million prospective voters were sent questionnaires.
  - The poll predicted that Landon would win.



# Bad sample: The *Literary Digest* case

- The prediction was a disaster: Roosevelt beat Landon in a landslide victory.
- *Literary Digest* magazine became a laughing stock.
- Magazine got shut down in 1938, never to return again.

What went wrong?



**SECRET BALLOT—No Signature—No Condition—No Obligation—Just Mark Your Choice—Mail at Once**

**CANDIDATES FOR PRESIDENT OFFICIALLY NOMINATED**  
(Names Arranged Alphabetically)

Put a Cross ☒ in Square Before the Name of Presidential Candidate You Prefer

|                                                        |                                                             |
|--------------------------------------------------------|-------------------------------------------------------------|
| <input type="checkbox"/> John W. Aldrich (Socialist)   | <input type="checkbox"/> William L. Dickinson (Unionist)    |
| <input type="checkbox"/> Earl Browder (Communist)      | <input type="checkbox"/> Franklin D. Roosevelt (Democratic) |
| <input type="checkbox"/> Leigh Colvin (Prohibitionist) | <input type="checkbox"/> Norman Thomas (Socialist)          |
| <input type="checkbox"/> Alfred M. Landon (Republican) | <input type="checkbox"/> _____                              |

**Mark How You Voted For President in 1932**

Did Not Vote: ☐ Under Legal Age ☐ Other Reasons ☐

To assist in tabulation please write name of your State here: \_\_\_\_\_

**FOR USE**

# The *Literary Digest* survey

## What went wrong?

- Potential voter names were compiled from sources such as **telephone directories** and **vehicle registration lists**.
  - These lists favored rich voters who leaned republican (Landon), leaving out poorer working-class citizens who supported Roosevelt.
- Many of the voters **did not respond**.
  - 2.3 million people actually responded. (still a pretty large sample)
  - Persons supporting Landon were **much more likely to respond** to the survey.
  - In fact, many Roosevelt supporters did not remember receiving a survey.

# Bad sample: The *Delphi-Facebook* survey

- What percentage of the US adult population is vaccinated against COVID-19?
- Survey done by Delphi Research Group at Carnegie Mellon University, in partnership with Facebook.

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Article | Published: 08 December 2021

## Unrepresentative big surveys significantly overestimated US vaccine uptake

[Valerie C. Bradley](#), [Shiro Kuriwaki](#), [Michael Isakov](#), [Dino Sejdinovic](#), [Xiao-Li Meng](#) & [Seth Flaxman](#) 

*Nature* **600**, 695–700 (2021) | [Cite this article](#)

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### Abstract

Surveys are a crucial tool for understanding public opinion and behaviour, and their accuracy depends on maintaining statistical representativeness of their target populations by minimizing biases from all sources. Increasing data size shrinks confidence intervals but

*“We show how a survey of 250,000 respondents can produce an estimate of the population mean that is **no more accurate than an estimate from a simple random sample of size 10**. Our central message is that **data quality matters more than data quantity**, and that compensating the former with the latter is a mathematically provable losing proposition.”*

- A massive 2021 study capturing roughly 250,000 weekly responses via Facebook.
- Facebook users opting into health surveys were **systematically more likely** to be vaccinated than the general public.
- Despite an enormous sample size, the survey **overestimated** actual US vaccine uptake by a massive 17 percentage points.
- The “Big data paradox.”

# Importance of *survey design*

More data  $\nRightarrow$  Better data

- Need to make sure that the sample selected is **representative** of the target population.
- A large **unrepresentative** sample may do **more damage** than a small one.
- Past accuracy of a flawed sampling procedure **does not guarantee** future results.
  - The *Literary Digest* surveys worked well for the previous five successive elections.
  - Need to have statistically sound sampling and estimation procedures.
  - Otherwise, disasters can happen anytime.

# Good sample



- A sample is **representative** if it can be used to “reconstruct” what the population looks like
  - and if we can provide an **accurate assessment** of how good that reconstruction is.
- Sample survey theory provides the only mathematically rigorous foundation for ensuring population representativeness.
- The **design** of the survey, i.e., **how the units are selected**, is very important.

# Good sample

## Objection to sampling

- “My opinion has **never been asked**, so how can the survey results claim to represent **me**?”
- “Extrapolating what tens of millions are thinking from a tiny sample of opinions **insults** human intelligence and **negates** true freedom of thought.”

## What can sampling buy us?

- **Reduced** cost, manpower, time, and **enlarged** scope.
- **Easier** to manage organizational work. Can use a small number of highly trained people.
- Estimates based on sample surveys are often **more accurate** than those based on a census.

$$\text{Total error} = \underbrace{\text{Sampling error}}_{\text{Error due to examining only a part of the population}} + \underbrace{\text{Nonsampling error}}_{\text{Measurement errors, missing data, poorly trained investigators}}$$

# Different types of sampling

## Probability sampling or Random sampling

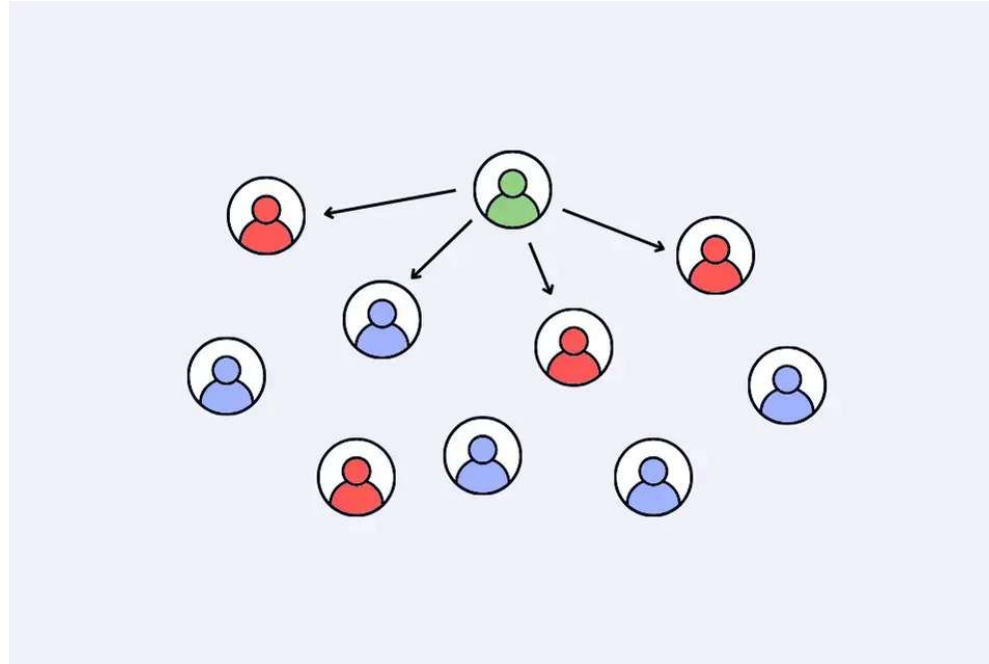
- The selection of units in the sample from a population is governed by a **mathematical randomization mechanism** (based on laws of chance or probability).
- Every unit in the sampling frame has a **known, strictly positive probability** of being selected.
- Can use this randomization mechanism to do **inference**.

## Non-probability sampling or Non-random sampling

- The selection of units in the sample from a population is **not governed** by the probability laws.
- **When it is used:** Exploratory research, qualitative studies, pilot testing, or situations where constructing a comprehensive sampling frame is impossible.
- Hard to do statistical inference.

# Non-random sampling: Some special cases

## Convenience sampling:



- Units are included simply because they are **immediately accessible and cheap to reach** for the researcher.
- **When it is used:** Quick pilot tests to check if survey wording makes sense.
- E.g., a researcher handing a draft questionnaire to people in the university canteen.

# Non-random sampling: Some special cases

## Judgment (or purposive) sampling:



- The researcher uses their own domain expertise to **hand-pick specific units** they believe will yield the most informative or relevant data.
- **When it is used:** Highly specialized case studies or expert interviews.
- E.g., a political analyst assessing a government's performance by intentionally interviewing only local “buddhijibis” (intellectuals).

# Non-random sampling: Some special cases

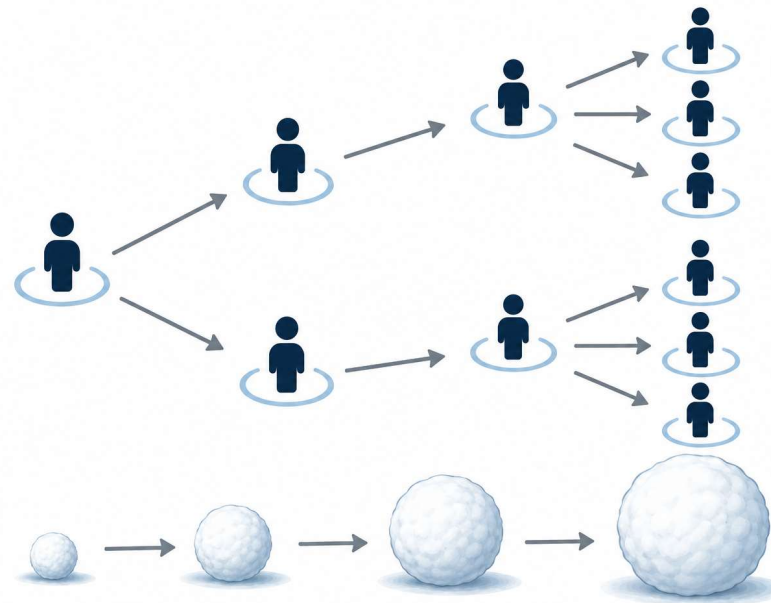
## Self-selection sampling:



- Individuals **voluntarily opt-in** to participate in the study entirely on their own initiative.
- **When it is used:** Customer feedback forms, informal internet/radio polls, or psychological studies requiring highly motivated volunteers.
- The statistics from such surveys **cannot be trusted**. At best, they are entertainment; at worst, they mislead.

# Non-random sampling: Some special cases

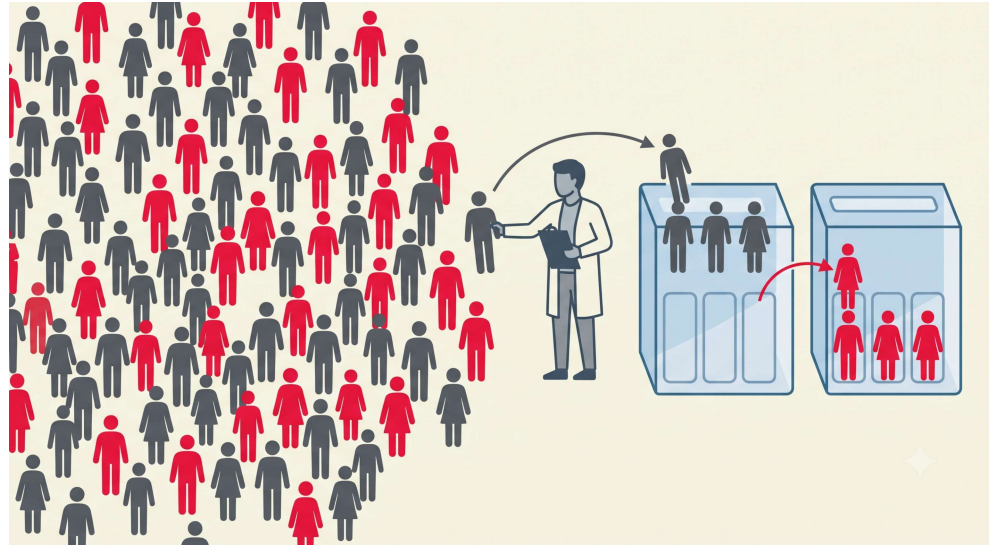
## Snowball sampling:



- The researcher selects one person (or a few people) for the study. These individuals are asked to **recruit or refer other individuals** from their personal networks who fit the study criteria.
- **When it is used:** Reaching hidden, rare, or vulnerable populations that lack a formal sampling frame.
- E.g., getting a list of patients having HIV/AIDS in a city may be difficult (because most of such persons won't want to be found). Here, snowball sampling can be used.

# Non-random sampling: Some special cases

## Quota sampling:



- The researcher sets a **specific target number** for different groups (like exactly 50 men and 50 women), and then fills those spots using convenience sampling until the target is met.
- **When it is used:** When you need a guaranteed mix of specific groups quickly and cheaply, without the effort or cost of actual random selection.
- E.g., a student wants feedback on the campus Wi-Fi. She sets a quota to interview exactly 25 first-year students and 25 second-year students and surveys students in the canteen until she hits the numbers.

# Sample Surveys

## Class 2

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Fall 2026

Previously  
on  
Sample Surveys...

# Recap

## Key components

- Observation unit, Target population
  - Mode of survey:
    - Complete enumeration or census, Sample survey
  - Sampled population, Sampled unit, Sampling frame
- 
- A good sample should be **representative** of the population.
  - The **design** of the survey, i.e., **how the units are selected**, is very important.

## Types of sampling

- Probability sampling or random sampling.
- Non-probability sampling or non-random sampling.
  - Convenience sampling
  - Judgment or purposive sampling
  - Self-selection sampling
  - Snowball sampling
  - Quota sampling

# Finite population sampling framework

## Running example

The state health ministry is planning to launch a new health insurance scheme. To budget for this, they must estimate the **average monthly out-of-pocket medical expenditure** for **all adults in West Bengal**.

## Setup

- **Finite** population of  $N$  units:  $U = \{1, 2, \dots, N\}$ .
  - Sampling frame.  $\rightarrow$  Units are labeled  $1, 2, \dots, N$ .
  - Population size:  $N$ .
- Study variable:  $Y$  (avg monthly out of pocket medical expenditure).
- $Y_j$  : The value of the study variable for unit  $j$  in the population.
- Parameter: some feature of the population based on  $Y$ 
  - Population mean:  $\bar{Y} = \frac{1}{N} \sum_{j=1}^N Y_j$ . (see the running example)
  - Population variance:  $\sigma^2 = \frac{1}{N} \sum_{j=1}^N (Y_j - \bar{Y})^2$ .
- In this setup, the  $Y_j$ s are treated as **fixed** (non-random) quantities.

# Finite population sampling framework

- $S$ : A random sample drawn from  $U$ .
- The units included in  $S$  are **random**.
- Sample size (number of units in the sample):  $n$ .
- The values of  $Y$  obtained in the sample:  $y_1, y_2, \dots, y_n$ .
  - For every  $i$ ,  $y_i$  is  $Y_j$  for some  $j \in U$ .
  - $y_1, y_2, \dots, y_n$  are **random variables**  $\rightarrow$  due to random sampling.
- $\mathcal{S}$  : Collection of all possible samples from  $U$ . (sample space)

## Sampling design

- A probability distribution over  $\mathcal{S}$ .
- $p : \mathcal{S} \rightarrow [0, 1]$ , such that  $\sum_{s \in \mathcal{S}} p(s) = 1$ .
- Example:  $U = \{1, 2, 3\}$ .  $\mathcal{S} = ?$ .
  - $p(\{1, 2\}) = p(\{1, 3\}) = p(\{2, 3\}) = \frac{1}{3}$ .
  - $p(s) = 0$ , otherwise.

# Simple Random Sampling (SRS)

- **Sampling scheme:** An algorithm/procedure to draw a sample.
- One such sampling scheme: Simple random sampling.
- The simplest form of probability sampling.
  - Select units one-by-one from the population.
  - At every stage of selection, every unit in the population has the **same marginal probability** of being selected.
- Sample size  $n$  is fixed.
- Two main types of SRS:

## Simple Random Sampling *With* Replacement (SRSWR)

- 1st draw: Select one of the  $N$  units with probability  $\frac{1}{N}$  each. That's the 1st unit in the sample.
- **Return** (i.e., place it back) it to the population.
- 2nd draw: Select one of the  $N$  units with probability  $\frac{1}{N}$  each. That's the 2nd unit in the sample. (**the same unit can be selected again**)
- Continue like this until  $n$  units are selected.
- At every stage, the marginal probability of selecting a particular unit of the population  $= \frac{1}{N}$ .

# Simple Random Sampling (SRS)

- **Sampling scheme:** An algorithm/procedure to draw a sample.
- One such sampling scheme: Simple random sampling.
- The simplest form of probability sampling.
  - Select units one-by-one from the population.
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- Sample size  $n$  is fixed.
- Two main types of SRS:

## Simple Random Sampling *Without* Replacement (SRSWOR)

- 1st draw: Select one of the  $N$  units with probability  $\frac{1}{N}$  each. That's the 1st unit in the sample.
- **Do not return** it to the population.
- 2nd draw: Select one of the **remaining**  $N - 1$  units with probability  $\frac{1}{N-1}$  each. That's the 2nd unit in the sample. (the same unit **cannot** be selected again)
- Continue like this until  $n$  units are selected.
- At every stage, the **marginal** probability of selecting a particular unit of the population  $= \frac{1}{N}$ . (why?)

# Implementation of SRS

Lottery method:



- Can be done with numbered paper folds in a box.
- Useful for small population size.
- **Troublesome** when the population size is large.

# Implementation of SRS

Random number table method:

|    | 00-04 | 05-09 | 10-14 | 15-19 | 20-24 | 25-29 | 30-34 | 35-39 | 40-44 | 45-49 |
|----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 00 | 54463 | 22662 | 65905 | 70639 | 79365 | 67382 | 29085 | 69831 | 47058 | 08186 |
| 01 | 15389 | 85205 | 18850 | 39226 | 42249 | 90669 | 96325 | 23248 | 60933 | 26927 |
| 02 | 85941 | 40756 | 82414 | 02015 | 13858 | 78030 | 16269 | 65978 | 01385 | 15345 |
| 03 | 61149 | 69440 | 11286 | 88218 | 58925 | 03638 | 52862 | 62733 | 33451 | 77455 |
| 04 | 05219 | 81619 | 10651 | 67079 | 92511 | 59888 | 84502 | 72095 | 83463 | 75577 |
| 05 | 41417 | 98326 | 87719 | 92294 | 46614 | 50948 | 64886 | 20002 | 97365 | 30976 |
| 06 | 28357 | 94070 | 20652 | 35774 | 16249 | 75019 | 21145 | 05217 | 47286 | 76305 |
| 07 | 17783 | 00015 | 10806 | 83091 | 91530 | 36466 | 39981 | 62481 | 49177 | 75779 |
| 08 | 40950 | 84820 | 29881 | 85966 | 62800 | 70326 | 84740 | 62660 | 77379 | 90279 |
| 09 | 82995 | 64157 | 66164 | 41180 | 10089 | 41757 | 78258 | 96488 | 88629 | 37231 |

- Proposed by Fisher and Yates. Available at the back of Statistics books.
- A Million Random Digits with 100,000 Normal Deviates* (1955), RAND Corporation.

## Example: SRSWR with $N = 10$ , $n = 4$

- Blindly start at any row of the table. Suppose we select **row 07**.
- Collect the  $n = 4$  **single-digit** numbers by moving left to right.  $\rightarrow 1, 7, 7, 8$ .
- Sample: Units 1,7,7,8.
- If we chose **row 08**, then the resulting sample: 4,10,9,5 (number 0  $\rightarrow$  unit 10).

# Implementation of SRS

Random number table method:

|    | 00-04 | 05-09 | 10-14 | 15-19 | 20-24 | 25-29 | 30-34 | 35-39 | 40-44 | 45-49 |
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Example: SRSWR with  $N = 17$ ,  $n = 4$

- Blindly start at any row of the table. Suppose we select row 07.
- Collect the  $n = 4$  double-digit numbers by moving left to right.
- Reject if a number is  $> 17$  or  $= 00$ . Sample: Units 17,15,10,03.
- **Issue:** High rejection rate.

# Implementation of SRS

Random number table method:

|    | 00-04 | 05-09 | 10-14 | 15-19 | 20-24 | 25-29 | 30-34 | 35-39 | 40-44 | 45-49 |
|----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
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Example: SRSWR with  $N = 17$ ,  $n = 4$

- **A fix:** Remainder method. Highest two-digit multiple of  $N = 17$  is  $17 \times 5 = 85$ .
- Reject if a number is  $> 85$  or  $= 00$ . If a number  $x$  is between 01 and 85, take  $x \bmod 17$ .
- If row 07 is chosen, then sample: units 17, 10, 13, 15.

# Implementation of SRS

- For SRSWOR, just ignore any repeated number and proceed.
- In R, super easy to draw both SRSWR and WOR samples.

```
# Define population and sample sizes
N <- 17
n <- 4

# Draw SRSWR
srswr_draw <- sample(1:N, size = n, replace = TRUE)

# Draw SRSWOR (replace = FALSE is default)
srswor_draw <- sample(1:N, size = n, replace = FALSE)
```

- Sampling from a given data frame (say, named pop\_data)

```
selected_indices <- sample(1:nrow(pop_data), size = n,
                           replace = TRUE)

sample_data <- pop_data[selected_indices, ]
```

# Simple Random Sampling *With* Replacement (SRSWR)

- Typically, an SRSWR sample, one retains the **order** of selected units and their **multiplicities**.
  - E.g., (4, 3, 2, 2, 8).
- What is the **sampling design** induced by SRSWR?

## SRSWR: Sampling design

- Set  $\mathcal{S}$ : set of all ordered samples of size  $n$  from  $U$ , allowing replacements.
- For an ordered sample  $\mathbf{s} \in \mathcal{S}$ ,

$$p(\mathbf{s}) = \frac{1}{N^n}.$$

- The sampling design is **uniform** over  $\mathcal{S}$ .

- Selection probability:  $\Pr(\text{Unit } j \text{ is selected in stage } k)$ 
  - Under SRSWR, the selection probability of each unit at each stage  $= \frac{1}{N}$ .
- Inclusion probability:  $\Pr(\text{Unit } j \text{ is included in the sample}) =: \pi_j$ .
  - Under SRSWR,  $\pi_j = 1 - (1 - \frac{1}{N})^n$ , for all  $j \in U$ .

# Estimation under SRSWR

- Parameter of interest: Population mean  $\bar{Y} = \frac{1}{N} \sum_{j=1}^N Y_j$ .
  - Sample of  $Y$  values:  $(y_1, y_2, \dots, y_n)$ .
  - What is a reasonable estimator of  $\bar{Y}$ ?
    - Sample mean:  $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ .
  - Bias of the estimator:  $\text{Bias}(\bar{y}) = \mathbb{E}(\bar{y}) - \bar{Y}$ .
  - Turns out,  $\text{Bias}(\bar{y}) = 0$ , i.e., sample mean is **unbiased** for the population mean under SRSWR.
- This result holds for any other types of population means, e.g.,  $\frac{1}{N} \sum_{j=1}^N h(Y_j)$ , with known  $h(\cdot)$ .
  - SRSWR sample is **representative** of the population. We can recover the population means **in expectation**.
  - **Importantly, no distributional assumptions needed** for unbiasedness. It holds **by design**.
  - All the results are similar to the case of i.i.d sampling from an infinite population.
- Need to provide a measure of accuracy of estimation.

# Estimation under SRSWR

- Population variance of  $Y$ :  $\sigma^2 = \frac{1}{N} \sum_{j=1}^N (Y_j - \bar{Y})^2$ .
- Turns out,  $\text{Var}(\bar{y}) = \frac{\sigma^2}{n}$ .  $\rightarrow$  Sampling variance of  $\bar{y}$ .
- But  $\sigma^2$  is **unknown**; it depends on  $Y_1, Y_2, \dots, Y_N$ .
  - Estimate it from the sample.
- Let  $s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$ .
- $\mathbb{E}(s^2) = \sigma^2$  (why?).
- An **unbiased** estimator of the sampling variance of  $\bar{y}$ :

$$\widehat{\text{Var}}(\bar{y}) = \frac{s^2}{n}.$$

# Simple Random Sampling *Without* Replacement (SRSWOR)

- Typically, an SRSWOR sample, one **gets rid of the order** of selected units.
  - E.g.,  $\{2, 3, 4, 7, 8\}$ .
- What is the **sampling design** induced by SRSWOR?

## SRSWOR: Sampling design

- Set  $\mathcal{S}$ : set of all unordered samples of size  $n$  from  $U$ , not allowing replacements.
- For an unordered sample  $s \in \mathcal{S}$ ,

$$p(s) = \frac{1}{\binom{N}{n}}.$$

- The sampling design is **uniform** over  $\mathcal{S}$ .

- Selection probability:  $\frac{1}{N}$  for all units, at every stage.
- Inclusion probability:
  - **First-order**:  $\pi_j = P(\text{Unit } j \text{ is included in the sample}) = \frac{n}{N}$  for all  $j \in U$ .
  - **Second-order**:  $\pi_{jj'} = P(\text{Units } j \text{ and } j' \text{ are included in the sample}) = \frac{n(n-1)}{N(N-1)}.$



# Sample Surveys

## Class 4

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Fall 2026

# Previously on Sample Surveys...

# Recap

- Estimation of population mean  $\bar{Y}$  under SRSWOR
  - Estimator: Sample mean  $\bar{y}$ .  $\bar{y}$  is **unbiased** for  $\bar{Y}$ .
  - **Sampling variance** of  $\bar{y}$ ,  $\text{Var}(\bar{y}) = (1 - f) \frac{s^2}{n}$ .
    - $1 - f$ : **Reduction in variability between samples** that happens due to different samples having some **common units**.
  - **Estimator** of sampling variance:  $\widehat{\text{Var}}(\bar{y}) = (1 - f) \frac{s^2}{n}$ .
- For large  $N$ , SRSWR: a quicker alternative to SRSWOR.
- Application of SRSWR: **Bootstrap** methods.
- Other estimands: Population total, population variance, population covariance (between two study variables).
- Estimating **population proportions**: Work with a binary  $Y$ , indicating the characteristic of interest.
- **Measures of accuracy**: sampling variance, standard errors, **coefficient of variation** or relative standard deviation.
- Another estimation method: SRSWR + use **distinct units** only.
  - Effective sample size:  $n_{\text{eff}}$
  - The sample mean  $\bar{y}_{\text{eff}}$  is unbiased for  $\bar{Y}$ .
  - **Is it better than usual SRSWR with  $n$  units?**

# Confidence interval

- **Interval estimation** of the parameter of interest.
- $100(1 - \alpha)\%$  confidence interval: A **random interval**  $[L, U]$  that contains the parameter of interest with probability at least  $1 - \alpha$ .

## Canonical case: IID sampling from Normal distribution

- $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$ ,  $\sigma$  known (say). Estimator of  $\mu$ :  $\hat{\mu} = \bar{X}$ .
  - $\bar{X} \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$  (sampling distribution).  $SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$ .
  - $100(1 - \alpha)\%$  confidence interval (CI) for  $\mu$ :  $[\bar{X} - z_{\frac{\alpha}{2}} \times SE(\bar{X}), \bar{X} + z_{\frac{\alpha}{2}} \times SE(\bar{X})]$ .
    - $z_{\frac{\alpha}{2}}$ :  $(1 - \frac{\alpha}{2})$ th quantile of  $\mathcal{N}(0, 1)$  distribution
  - For  $\alpha = 0.05$ ,  $z_{\frac{\alpha}{2}} \approx 1.96$ .
  - 95% CI for  $\mu$ :  $[\bar{X} - 1.96 \times \frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96 \times \frac{\sigma}{\sqrt{n}}]$ .
  - **Coverage probability** of the above random interval: 0.95.
- 
- 95% CI for  $\bar{Y}$  in finite population sampling:
    - If we independently draw many random samples from the finite population, at least 95% of the resulting random intervals will contain the true value of  $\bar{Y}$ .

# Confidence interval in finite population sampling

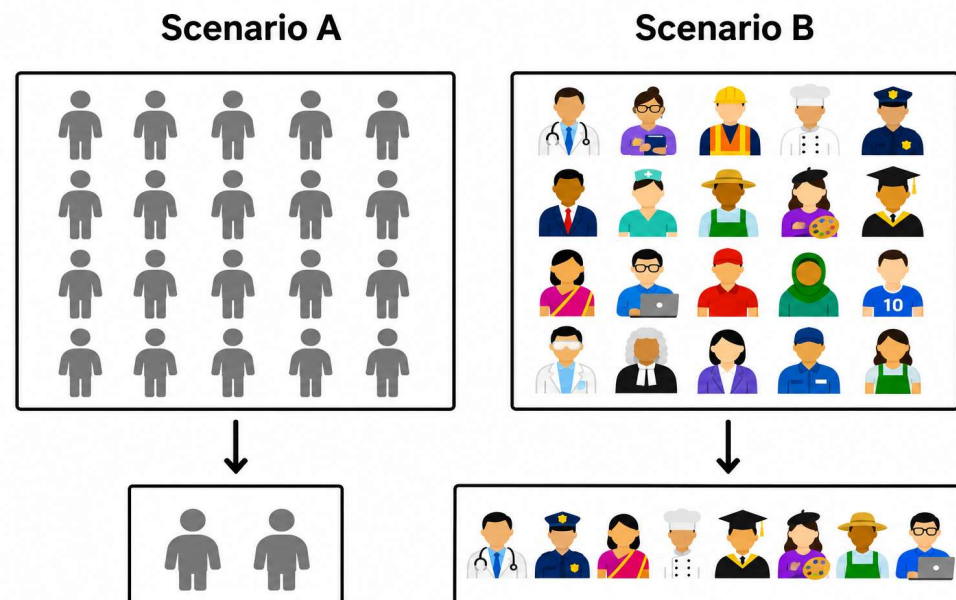
- In SRS, the sampling distribution of  $\bar{y}$  depends on  $Y_1, \dots, Y_N$  (**unknown**).
  - If we knew the exact sampling distribution, we would be able to compute the exact coverage probability of any random interval.
- **Cannot get an exact 95% CI for  $\bar{Y}$  based on  $\bar{y}$ .**
- Need to rely on **asymptotic approximations**.
- **Rough idea:** Hájek (1960) showed that, under some **regularity conditions** on the finite population, and if  $N$ ,  $n$ , and  $(N - n)$  are **sufficiently large**,

$$\frac{\bar{y} - \bar{Y}}{SE(\bar{y})} \underset{\text{approx}}{\sim} \mathcal{N}(0, 1).$$

- That is, sampling distribution of  $\bar{y}$  can be approximated by a Normal distribution.
- **Approximately**,  $[\bar{y} - 1.96 \times SE(\bar{y}), \bar{y} + 1.96 \times SE(\bar{y})]$  contains the population mean with probability 0.95.
- Plug in the estimate of the standard error.
- Approx 95% CI for  $\bar{Y}$  :  $[\bar{y} - 1.96 \times \widehat{SE}(\bar{y}), \bar{y} + 1.96 \times \widehat{SE}(\bar{y})]$ .
- For small  $n$ ,  $t_{n-1}$ -distribution gives a better approximation to the sampling distribution.

# How to choose the sample size $n$ ?

- One of the most important practical problems in sample surveys.
  - Too small  $n$ : Estimate may be completely **unreliable**.
  - Too large  $n$ : Very **costly** and **time consuming**. Often you have a limited budget.
  - “Bigger is always better” – not always true (**non-sampling errors**).
  - What  $n$  is **just right**?
- Assume no non-sampling errors for now.
- **No one size fits all formula.**



# Determination of $n$ : a recipe

- Ask yourself: What do I want to achieve with my sample?
  - How much precision do I need?
  - How much error will I tolerate?
- Don't think about: what percentage of the population should I include in my sample.
- Find an equation that relates  $n$  with your expectations of the sample.
- Solve for  $n$ . (not straightforward though)
- If the resulting  $n$  is much larger than you can afford, adjust your expectations.
- If you can't even come close to the precision you want with the resources you have, think whether you should even conduct the survey.

# Margin of error

- Determines how much precision is needed and how much error is tolerable.
- In absolute terms:  $P(|\bar{y} - \bar{Y}| \leq e) = 1 - \alpha$ .
  - $e$  : Absolute margin of error.
- In relative terms:  $P\left(\left|\frac{\bar{y} - \bar{Y}}{\bar{Y}}\right| \leq r\right) = 1 - \alpha$ .
  - $r$  : Relative margin of error.
- You specify  $e$  (or  $r$ ), and find an appropriate  $n$ .
  - Connect to the approximate  $100(1 - \alpha)\%$  CI.
- What if we don't specify a margin of error?
  - E.g., we want to estimate the average monthly rent for a Paying Guest accommodation in Baranagar. Suppose we set  $n = 15$ .
  - Get CI: ₹6000 ± ₹1265, i.e., [₹4735, ₹7265].
  - What we should do instead: Set  $e$ , e.g.,  $e = ₹300$ .
- Another example: Suppose we want to estimate the mean birth weight of infants born to mothers who smoke cigarettes during pregnancy.
  - Margin of error should be pretty small, i.e., 0.9 kg.

# Determining $n$ for SRS

- WLOG, focus on absolute margin of error  $e$ .
- For SRSWR, solving for  $n$ , we get:  $n_{WR} = \frac{z_{\frac{\alpha}{2}}^2 \sigma^2}{e^2}$ .
- For SRSWOR, solving for  $n$ , we get:  $n_{WOR} = \frac{z_{\frac{\alpha}{2}}^2 S^2}{e^2 + \frac{z_{\frac{\alpha}{2}}^2 S^2}{N}}$ .
- $n_{WOR} \approx \frac{n_{WR}}{1 + \frac{n_{WR}}{N}}$ .
- But  $n_{WR}$  and  $n_{WOR}$  depend on **unknown population quantities**.

## Special case: Estimation of population proportion

- Parameter of interest:  $P$ .
- $n_{WR} = \frac{z_{\frac{\alpha}{2}}^2 P(1-P)}{e^2}$ ,  $n_{WOR} \approx \frac{z_{\frac{\alpha}{2}}^2 P(1-P)}{e^2 + \frac{z_{\frac{\alpha}{2}}^2 P(1-P)}{N}}$ .
- Note that  $P(1-P) \leq \frac{1}{4}$ . Put it back in the formula.
- Get  $n$  corresponding to the **worst-case** scenario.

# Sample Surveys

## Class 5

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# Previously on Sample Surveys...

# Recap

- Confidence intervals (CI) for  $\bar{Y}$ .
  - Exact CI **cannot** be computed.
  - Rely on **asymptotic approximations**.
  - Under some conditions, and for large  $N, n, N - n$ , the sampling distribution of  $\bar{y}$  can be approximated by a **Normal** distribution.
  - For SRS, approximate 95% CI for  $\bar{Y}$  is of the form

$$[\bar{y} - 1.96 \times \widehat{SE}(\bar{y}), \bar{y} + 1.96 \times \widehat{SE}(\bar{y})]$$

- Sample size determination.
  - **Margin of error**:  $e$  (absolute) and  $r$  (relative).
  - Connect  $n$  with the margin of error via some equation. → **equate with CI for  $\bar{Y}$** .
  - For SRSWR,  $n_{WR} = \frac{z_{\frac{\alpha}{2}}^2 \sigma^2}{e^2}$ .
  - For SRSWOR,  $n_{WOR} = \frac{z_{\frac{\alpha}{2}}^2 S^2}{e^2 + \frac{z_{\frac{\alpha}{2}}^2}{N}} \approx \frac{n_{WR}}{1 + \frac{n_{WR}}{N}}$ .
  - Desired choice depends on **unknown** population quantities.

# Choosing $n$ for estimating population proportions

- Parameter of interest:  $P$ .
- $n_{WR} = \frac{z_{\frac{\alpha}{2}}^2 P(1-P)}{e^2}$ ,  $n_{WOR} \approx \frac{z_{\frac{\alpha}{2}}^2 P(1-P)}{e^2 + \frac{z_{\frac{\alpha}{2}}^2 P(1-P)}{N}}$ .
- **Worst-case approach:** When the population is the **most unpredictable**.
  - Assume the state of **maximum uncertainty**, which requires the **largest sample size** to cut through the noise.
  - Corresponds to  $P = 0.5$ . Get  $n$  for this choice of  $P$ .
  - If true  $P$  is 0.5, we will have surveyed **exactly enough** people.
  - If true  $P$  is something else (e.g., 0.20), our  $n$  will be **larger than strictly necessary**.

## Worst-case approach: an example

- What proportion of the population uses net banking on their mobile phones?
- Set margin of error  $e = 0.05$ .
- $n_{WR} = \frac{1.96^2 \times 0.25}{0.05^2} = 384.16$ . Choose  $n$  to be 385.
- If we make  $e$  **higher**, e.g.,  $e = 0.1$  resulting  $n$  will be **lower**, i.e.,  $n = 97$ .
- In general, what if we focus on **relative** margin of error  $r$ ?

# Determining $n$ for SRS: general guidelines

- With absolute error, the resulting  $n$  depends on  $S^2$  (or  $\sigma^2$ ).
- **How to plug in?**
  - Bound by the worst-case value (e.g., as in the population proportion case).
  - **Pilot survey:** Do a **small survey before the actual survey** to get initial estimates of these parameters.
  - Use previous studies or data available in the literature.
  - If nothing else is available, **guess the variance**. Use a theoretical model if needed.
- With relative error, the resulting  $n$  depends on the **CV** of  $Y$  in the population.
- **How to plug in?**
  - Similar ideas as before.
  - Interestingly, CV tends to be **more stable** over time and is **easier to guess** in advance. (e.g., house prices)
- We often have **multiple study variables**.
  - Different variables demand different sample sizes.
  - Focus on one or two important variables (a compromise).

# Choice of $n$ : a hypothesis testing perspective

- Previous approach: estimation perspective.
  - Trap the true parameter within a margin of error with high probability.
  - Choice of  $n$  is driven by  $e$  (or  $r$ ) and  $\alpha$  (confidence level).

## Hypothesis testing approach

- Goal: To make a binary, actionable **decision** about the unknown parameter.
  - E.g., Is the true average rent in the city strictly greater than ₹6,000?
  - To test  $H_0 : \bar{Y} = 6000$  vs  $H_1 : \bar{Y} > 6000$ .
  - Choice of  $n$  is driven by two distinct types of **decision making risks**.
    - $\alpha = P(\text{Type-I error})$ .  $\rightarrow$  The risk of a **false positive**.
    - $\beta = P(\text{Type-II error})$ .  $\rightarrow$  The risk of a **false negative**.
- Power of the test:  $1 - \beta$ .
- Focus on simple null vs simple alternative.

$$H_0 : \bar{Y} = \mu_0 \quad \text{vs} \quad H_1 : \bar{Y} = \mu_1 (> \mu_0).$$

*If the true mean is actually  $\mu_1$ , I want to be **confident with probability**  $1 - \beta$  that my test will catch it. **What sample size do I need?***

# Choice of $n$ : a hypothesis testing perspective

- Testing problem:  $H_0 : \bar{Y} = \mu_0$  vs  $H_1 : \bar{Y} = \mu_1 (> \mu_0)$ .
- **Effect size**:  $\Delta = \mu_1 - \mu_0$ .  $\rightarrow$  Helps to choose the relevant  $\mu_1$ .
  - **Minimum detectable effect**: How big does the shift need to be before we actually care enough to do something about it?
  - House rent example: Maybe we only care if the rent has jumped by at least ₹500. Choose  $\Delta = 500$ .

## Example: Worker strike

- The workers' union in a food delivery company is thinking about going on strike to demand a higher salary due to a potential increase in fuel expense.
- **Baseline**: Historical data shows the average daily fuel expense is ₹150 ( $= \mu_0$ ).
- **Tipping point**: A ₹50 increase wipes out the exact payout of one average food delivery. So,  $\Delta = 50$ , hence  $\mu_1 = 200$ .

## Example: Manufacturing and quality control ( $\mu_1 < \mu_0$ case)

- A factory produces steel bolts for bridges. The baseline tensile strength ( $\mu_0$ ) is 500 megapascals (MPa). They suspect the new batch of steel may be weaker.
- If the strength drops to 490 MPa, the bridge becomes unsafe. So,  $\mu_1 = 490$ .

# Choice of $n$ for testing: A general recipe

- Set your effect size (and hence the alternative hypothesis).
  - Set the values of the probability of Type-I error and Power. E.g.,  $\alpha = 0.05$ ,  $1 - \beta = 0.8$ .
  - Create an equation connecting  $n$  to the above quantities and solve for  $n$ .
  - Check if sampling with this choice of  $n$  is feasible given the budget. If not, **revise your criterion**.
    - Increase  $\alpha$ : we increase the chance of accidentally claiming a shift happened **when it actually hasn't**. (rarely done though)
    - Accept a higher risk of **missing the shift**: Increase  $\beta$ , i.e., lower the Power.
    - Accept only caring about **larger, more obvious** changes: Increase  $\Delta$ .
- 
- $H_0 : \bar{Y} = \mu_0$  vs  $H_1 : \bar{Y} = \mu_1 (> \mu_0)$
  - Consider a right-tailed test. **How to perform the test?**
  - **Asymptotic approximations**:  $\frac{\bar{y} - \bar{Y}}{SE(\bar{y})} \overset{\text{approx}}{\sim} \mathcal{N}(0, 1)$ .
  - **Pretend** for now that you know  $SE(\bar{y})$ .
  - Test statistic:  $t = \frac{\bar{y} - \mu_0}{SE(\bar{y})}$ . Reject  $H_0$  at level  $\alpha$  if  $t > z_\alpha$ .
    - $z_\alpha$  is the  $(1 - \alpha)$ th quantile of a standard Normal distribution.

# SRS: finding the right $n$ for testing

- Compute the **Power of the test**, and solve for  $n$ .
- For SRSWR,  $n_{WR} = \frac{(z_{\alpha} + z_{\beta})^2 \sigma^2}{\Delta^2}$ .
- For SRSWOR,  $n_{WOR} = \frac{(z_{\alpha} + z_{\beta})^2 S^2}{\Delta^2 + \frac{(z_{\alpha} + z_{\beta})^2 S^2}{N}}$ . So,  $n_{WOR} \approx \frac{n_{WR}}{1 + \frac{n_{WR}}{N}}$ . (same as before)
- For two sided tests,  $n_{WR} \approx \frac{(z_{\frac{\alpha}{2}} + z_{\beta})^2 \sigma^2}{\Delta^2}$ ,  $n_{WOR} \approx \frac{n_{WR}}{1 + \frac{n_{WR}}{N}}$ .
- **Plug in  $\sigma^2$** : Similar idea as before.

## Example: Diabetes case study

- An investigator hypothesizes that in people free of diabetes, fasting blood glucose is **higher** in those who drink at least 2 cups of coffee per day.
- Historical data: mean fasting blood glucose level in people free of diabetes is reported as 95.0 mg/dL with an sd of 9.8 mg/dL.
- If the mean blood glucose level in people who drink at least 2 cups of coffee per day is 100 mg/dL, this would be important clinically.
- How many patients should be enrolled in the study (under SRSWR) to ensure that the power of a two-sided test at 5% level is 80% to detect this difference?
- $n_{WR} = 30.185$ . Set  $n = 31$ .

# Estimating population proportions: choice of $n$

- $H_0 : P = P_0$  vs  $H_1 : P = P_1(> P_0)$ .
- Consider a right-tailed test.
- For SRSWR,  $n_{WR} = \frac{(z_\alpha + z_\beta)^2 P_0(1-P_0)}{\Delta^2}$ .  $\rightarrow$  everything is known.
- For SRSWOR,  $n_{WOR} \approx \frac{n_{WR}}{1 + \frac{n_{WR}}{N}}$ . (same as before)
- Two-sided tests: similar approximations as before.

## Example: Heart disease case study

- A recent report from the Framingham Heart Study indicated that 26% of people free of heart disease had elevated LDL cholesterol levels, defined as LDL > 159 mg/dL.
- An investigator hypothesizes that, among patients with a history of heart disease, a higher proportion will have elevated LDL cholesterol.
- How many patients should be studied to ensure that the power of a two-sided test at 5% level is 90% to detect a 0.05 difference in the proportion with elevated LDL cholesterol?
- $n_{WR} = 808.75$ . Set  $n = 809$ .

# Sample Surveys

## Class 6

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Fall 2026

# Previously on Sample Surveys...

# Recap

- Sample size determination: **estimation** perspective
- Hypothesis testing for  $\bar{Y}$ .
  - Exact test **cannot** be performed.
  - Rely on **asymptotic approximations**.
- Sample size determination: **testing** perspective
  - **Effect size**:  $\Delta$ .
  - **Level of significance**:  $\alpha$
  - **Power**:  $1 - \beta$ .
  - Type of the test (one-sided, two-sided)
- Consider two-sided tests.
- For SRSWR,  $n_{WR} \approx \frac{(z_{\frac{\alpha}{2}} + z_{\beta})^2 \sigma^2}{\Delta^2}$ 
  - For estimation problem with margin of error  $e$ ,  $n_{WR} = \frac{z_{\frac{\alpha}{2}}^2 \sigma^2}{e^2}$ .
- For SRSWOR,  $n_{WOR} \approx \frac{n_{WR}}{1 + \frac{n_{WR}}{N}}$ .
- Desired choice depends on **unknown** population quantities.
  - Plug-in initial estimates: **Pilot survey**.
  - Use upper-bounds: **Worst-case approach**.

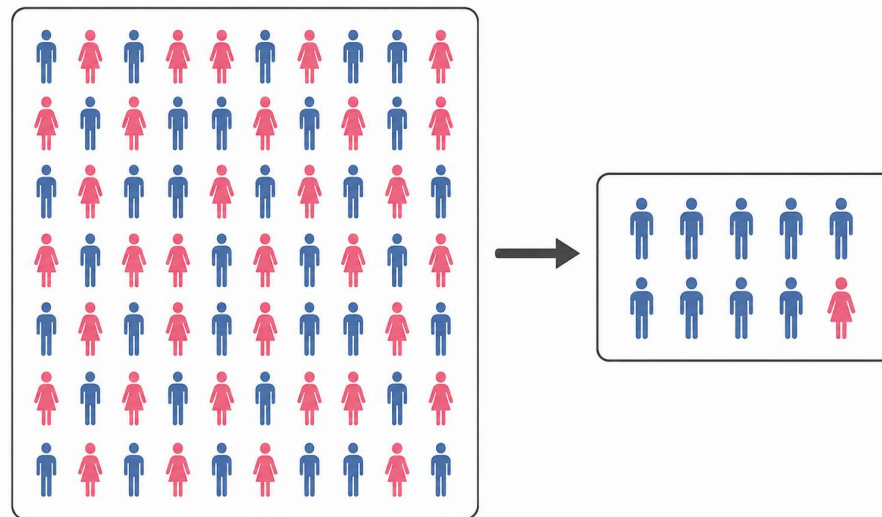
# Simple random sampling: advantages

- Conceptually simple. Easy to explain to non-statisticians.
- Eliminates researcher's bias.
- Easy to implement in most cases.
  - You only need the complete list of population units, nothing else.
- Facilitates inference for many population parameters of interest.
  - Unbiased estimation.
  - Under pretty mild conditions, obtain confidence intervals and perform hypothesis tests.
  - Methodologically, very similar to familiar iid sampling.
- Can something go wrong?
- Can we do better?

# An example

## Campus health survey

- A university health center wants to estimate the average height of the 1,000 undergraduate students on campus.
- There are 50% male students and 50% female students.
- We perform an SRSWOR.

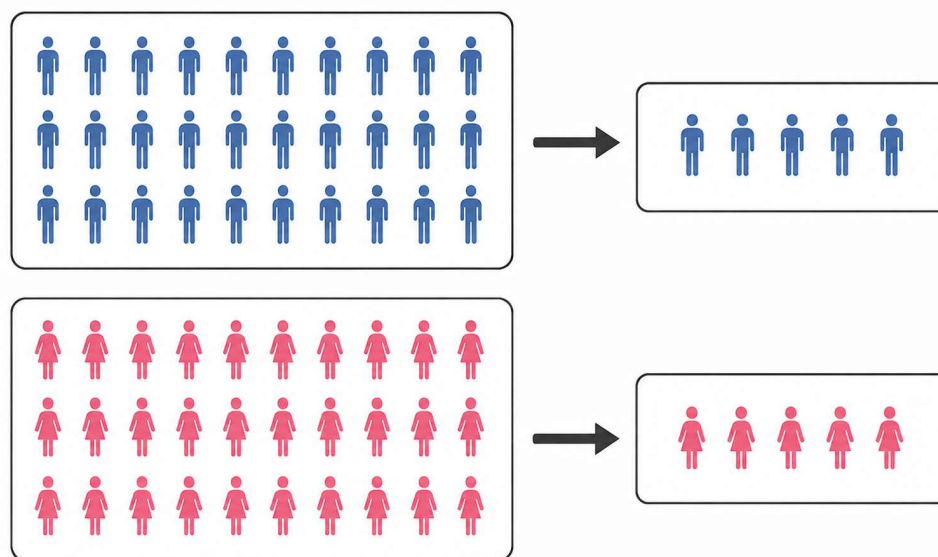


- What went wrong?
  - SRS gives representativeness **in expectation**.
  - Accidentally, we can get a highly **unrepresentative** sample.

# An example

- Suppose we have information on the gender of the units in the population. → **auxiliary data**.
- SRS is completely blind to this information.

A fix:



- Still a **random sample**.
- A much **more representative** sample than the previous one (in terms of gender).
- An example of **Stratified random sampling**.
- What are the statistical implications?

# Stratified random sampling

- *Stratify*: To arrange or deposit in layers.
- Divide the population into a few disjoint subpopulations, called **strata**.
  - Singular: Stratum, Plural: Strata.
- Assume: We know the stratum membership of every unit in the population.

## Stratified random sampling

Draw independent simple random samples from each stratum.

- Directly uses **auxiliary information** in the sampling process.

## Setup

- $H$ : Number of strata. Index the strata by  $h = 1, 2, \dots, H$ .
- $N_h$ : Size of stratum  $h$  in the population.
  - $N_1 + N_2 + \dots + N_H = N$ .
- $n_h$ : Size of the SRS to be drawn from stratum  $h$ . ( $n_h \geq 1$ )
  - $n_1 + n_2 + \dots + n_H = n$ .
- We focus on sampling **without replacement**.

# Stratified sampling: Estimation of $\bar{Y}$

- Use double indexing:  $h$  for stratum,  $j$  for unit.
- $Y_{hj}$  : Value of the study variable for unit  $j$  in stratum  $h$ .
- $\bar{Y}_h = \frac{1}{N_h} \sum_{j=1}^{N_h} Y_{hj}$  Mean of  $Y$  in stratum  $h$ .
- $\bar{Y} = \frac{1}{N} \sum_{h=1}^H N_h \bar{Y}_h$ .

## What is a reasonable estimator of $\bar{Y}$

- This is not an **equal probability** sample in general.
  - $\bar{y}$  **may not** be unbiased.
  - Let  $\bar{y}_h = \frac{1}{n_h} \sum_{i=1}^{n_h} y_{hi}$ : Sample mean in stratum  $h$ .
  - $\bar{y}_{st} = \frac{1}{N} \sum_{h=1}^H N_h \bar{y}_h$ .
    - $\mathbb{E}(\bar{y}_{st}) = \bar{Y}$ . (**unbiased**)
- 
- What about its variance? How to estimate it?

# Stratified sampling: Estimation of $\bar{Y}$

- $\text{Var}(\bar{y}_{st}) = \frac{1}{N^2} \sum_{h=1}^H N_h^2 (1 - f_h) \frac{S_h^2}{n_h}$ .
  - Same notations as before, with subscript  $h$  to indicate the stratum.
- $\widehat{\text{Var}}(\bar{y}_{st}) = \frac{1}{N^2} \sum_{h=1}^H N_h^2 (1 - f_h) \frac{s_h^2}{n_h}$ .
- When is this variance **small**? Is it **better** than usual SRSWOR of size  $n$ ?
  - Units within each stratum should be **as homogeneous as possible**.
- Approximate  $100(1 - \alpha)\%$  CI for  $\bar{Y}$ :  $[\bar{y}_{st} - z_{\frac{\alpha}{2}} \text{SE}(\bar{y}_{st}), \bar{y}_{st} + z_{\frac{\alpha}{2}} \text{SE}(\bar{y}_{st})]$ .
- Testing for  $\bar{Y}$ : Similar to the SRSWOR case.

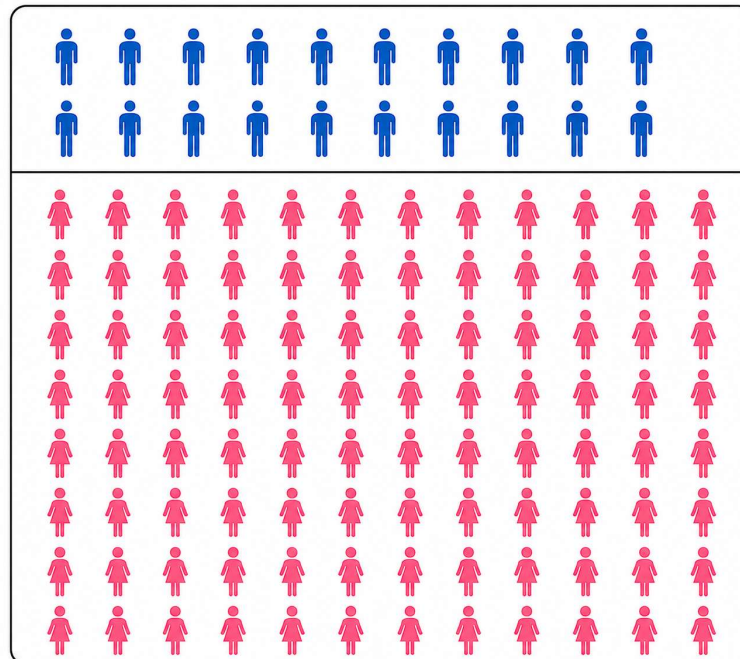
# Strata-wise allocation of sample sizes

- Suppose  $n$  is given. We want to decide how to allocate it to the  $H$  strata.
- I.e., what are the choices of  $n_1, n_2, \dots, n_H$ ?

## Equal allocation

$$n_h = \frac{n}{H} \text{ for all } h \in \{1, 2, \dots, H\}.$$

Is it reasonable in the following scenario?



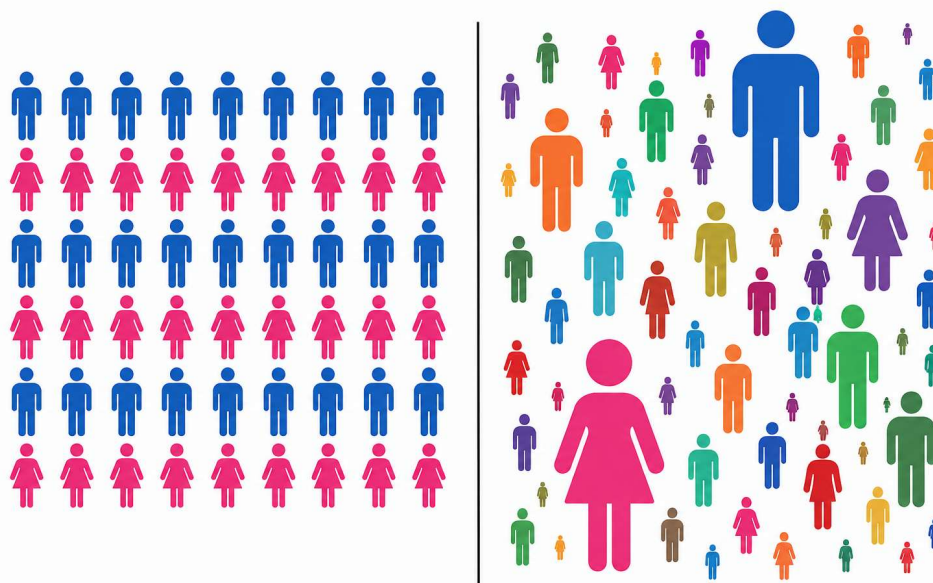
# Choice of $n_h$ : Proportional allocation

## Proportional allocation

- $n_h \propto N_h \rightarrow$  sample size is **proportional** to the population size within each stratum.
  - $n_h = \frac{n}{N} N_h$ .
  - Most frequently used.
  - If  $N_h$ s are equal, back to equal allocation.
- 
- Consider two designs, and two corresponding estimators of  $\bar{Y}$ :
    - 1 SRSWOR of size  $n$ . Use  $\bar{y}$ .
    - 2 Stratified random sampling WOR of size  $n$ , with proportional allocation. Use  $\bar{y}_{st}$ .
  - Which one is more efficient?
  - $\text{Var}_{srs}(\bar{y}) \geq \text{Var}_{st}(\bar{y}_{st})$ .
    - With **same** overall sample size  $n$ , stratification gives **better precision**.
    - Equality case?

# Choice of $n_h$ : Proportional allocation

- Suppose you want to estimate the average monthly income of a group of 1,000 people currently attending a conference.
- Two strata:
  - **500 PhD Scholars:** On the exact same fixed government fellowship (e.g., exactly ₹35,000 per month).
  - **500 Industry Alumni:** Some are currently between jobs (₹0/month), some are junior engineers (₹50,000/month), and a few are tech CEOs earning ₹10 Lakhs a month.



# Sample Surveys

## Class 7

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Fall 2026

# Previously on Sample Surveys...

# Recap

- Simple random sampling.
  - May lead to **bad samples** → not exploring different parts of the population well enough.
  - **Does not** utilize auxiliary information.
  - Scope of improvement: in terms of **precision**.
- **Stratified random sampling:**
  - Divide the population into  $H$  strata. Perform SRS from each stratum **independently**.
  - Uses auxiliary information. Expected to **scan** the population better.
- Sampling design?
  - 1st and 2nd order inclusion probabilities.
- Estimation of  $\bar{Y}$ :
  - $\bar{y}_{st} = \frac{1}{N} \sum_{h=1}^H N_h \bar{y}_h$ , where  $\bar{y}_h$ 's are stratum-wise means.
  - Variance estimation, CI, testing: similar.
  - Want each stratum to be **homogeneous**: small variability of  $Y$  **within** each stratum. (WSS)
  - Want high variability of  $Y$  **across** different strata. (BSS)
- Allocation of sample size into different strata.
  - Equal allocation, Proportional allocation.

# Comparison of variances

- Consider two designs:
  - SRSWOR of size  $n$ .
  - Stratified random sampling (WOR) of size  $n$  with **proportional allocation**.
- Denote  $W_h = \frac{N_h}{N}$ .  $\rightarrow$  **Population strength** of stratum  $h$ .
- $\bar{y}_{st} = \sum_{h=1}^H W_h \bar{y}_h$ .
- Without any large  $N$ -based approximations:

$$\text{Var}_{\text{SRS}}(\bar{y}) - \text{Var}_{st}(\bar{y}_{st}) = \frac{N-n}{n(N-1)} \left[ \sum_{h=1}^H W_h (\bar{Y}_h - \bar{Y})^2 - \frac{1}{N} \sum_{h=1}^H (1 - W_h) S_h^2 \right]$$

## Which sampling design is better?

- Small within-group variation, large across/between group variation  $\implies$  **stratification helps**.
- Special case:  $\bar{Y}_1 = \bar{Y}_2 = \dots = \bar{Y}_H$  (no variation in means across strata).
  - **SRSWOR is better**.
  - What went wrong?  $\rightarrow$  Strata were formed **poorly**.
- Realistic cases: **Stratification is always beneficial**.

# Choice of $n_h$ : Neyman allocation

- Proportional allocation takes into account the **sizes** of the strata.
- Neyman allocation takes into account both the **sizes** and the **within variability** of the strata.

## Neyman allocation

- $n_h \propto N_h S_h$ .
- $n_h = \frac{n N_h S_h}{\sum_{h'=1}^H N_{h'} S_{h'}}$  (from the overall sample size requirement).
- More difficult to implement than proportional allocation:  $S_h$  values are **unknown**.
  - Plug in initial estimates from a pilot survey.
  - Knowledge about the **relative values** of  $S_h$ .
- Neyman allocation vs Proportional allocation. Which one is more efficient?
- $\text{Var}_{st\_prop}(\bar{y}_{st}) \geq \text{Var}_{st\_Neyman}(\bar{y}_{st})$ .
- Equality case?

Neyman allocation **minimizes** the variance of  $\bar{y}_{st}$  with respect to  $n_1, \dots, n_H$ , subject to  $\sum_{h=1}^H n_h = n$ .

# Choice of $n_h$ : Optimal allocation with cost constraint

- Takes into account the **cost** of sampling.
- $A$ : **Overhead cost** of conducting the survey.
  - E.g., cost of conducting several meetings, data processing cost.
- $C_h$  : Cost of collecting info from a unit in strata  $h$ .
- Total cost:  $A + \sum_{h=1}^H n_h C_h$ .

## Optimization problem

$$\begin{aligned} \min_{n_1, \dots, n_H} \quad & \text{Var}(y_{st}) \\ \text{subject to} \quad & A + \sum_{h=1}^H n_h C_h \leq C. \end{aligned}$$

- $n_h \propto \frac{N_h S_h}{\sqrt{C_h}}$ .
- $n_h = \frac{(C-A) \frac{N_h S_h}{\sqrt{C_h}}}{\sum_{h'=1}^H \frac{N_{h'} S_{h'}}{\sqrt{C_{h'}}}}$  (from cost constraint).
- Equal stratum-wise cost (i.e.,  $C_h = \text{constant}$ )  $\implies$  Neyman allocation.

# Sample Surveys

Class 8

Ambarish Chattopadhyay

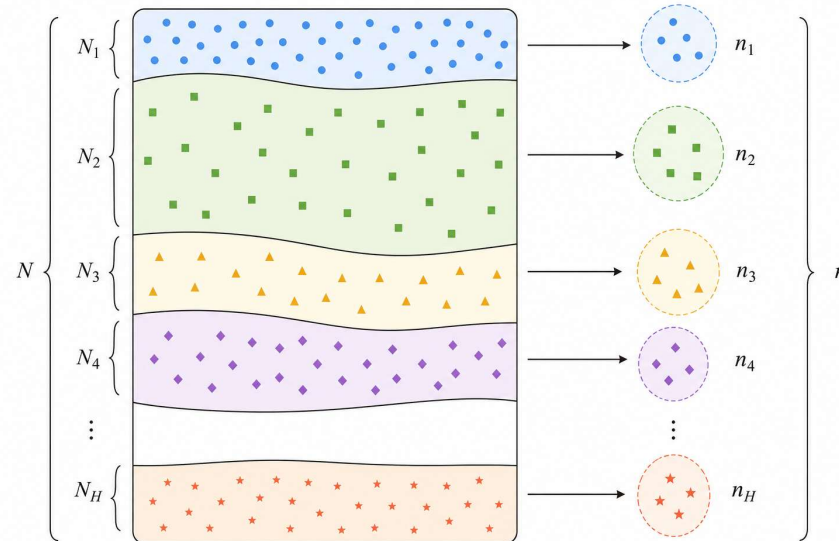
Indian Statistical Institute, Kolkata

Fall 2026

# Previously on Sample Surveys...

# Recap

Stratified random sampling:



## Allocation of sample size into different strata

- Proportional allocation:  $n_h \propto N_h$ .
- Neyman allocation:  $n_h \propto N_h S_h$ .
- Optimal allocation with cost constraints:  $n_h \propto \frac{N_h S_h}{\sqrt{C_h}}$ .

- Proportional allocation vs usual SRSWOR.
- Neyman allocation vs proportional allocation.

# Proportional and Neyman allocation

## Example

- $N = 200$  units, divided into  $H = 3$  strata.
    - Estimating monthly electricity usage for 200 city buildings. We group them into Factories, Retail Stores, and Small Offices.
  - $N_1 = 60, N_2 = 90, N_3 = 50$ .
  - $S_1 = 2S_2 = 4S_3$ .
  - $n = 30$ .
  - Proportional allocation:  $n_{1,\text{prop}} = 9, n_{2,\text{prop}} = 13.5, n_{3,\text{prop}} = 7.5$ .
  - Neyman allocation:  $n_{1,\text{Neyman}} = 15.3, n_{2,\text{Neyman}} = 11.5, n_{3,\text{Neyman}} = 3.2$ .
  - How to round off?
- 
- One solution: Change the domain of the optimization problem to integers only.
    - Wright (2017) proposes a greedy algorithm to solve this.
    - R-package allocation.
  - Brito et al. (2015): Multiple study variables. Formulates everything into a single integer programming problem.

# Optimal allocation

## Example

- $N = 15$  units, divided into  $H = 2$  strata.
    - A city has 15 water treatment stations, and we want to run a chemical audit on them. We group them into Old Stations and Modern Stations.
  - $N_1 = 5$ ,  $N_2 = 10$ .
  - $S_1 = 16$ ,  $S_2 = 2$ .
  - $n = 10$ .
  - Assume: Cost of survey is the same for all 15 units.
  - Optimal allocation:  $n_{1,\text{opt}} = 8$ ,  $n_{2,\text{opt}} = 2$ .
  - Here,  $n_{1,\text{opt}}$  exceeds  $N_1 \rightarrow$  oversampling boundary problem.
- 
- Classic method: Sample everyone from such a stratum. Allocate the extra sample size to the remaining strata.
  - Modern method: Wesołowski et al. (2024): introduces box-constraints on the sample sizes.
    - R-package stratallo.

# Stratification: some remarks

## Why should we use stratified random sampling?

- A good stratification protects us from getting a really bad sample.
- Stratified sampling often gives more precise estimates than SRS.
- Stratification allows us to estimate **strata-level parameters**.
  - Average income within each district.
  - Proportion of infected people among males and females.
- A stratified sample may be **more convenient** to administer, i.e., lower overall cost.
- **Sampling frames** may be constructed differently in different strata.
  - E.g., using a corporate database to sample big companies, but using street maps to sample local vendors.
- Different **field procedures** may be used for different strata.
  - E.g., emailing online surveys to hospital staff, but doing face-to-face interviews with rural patients.
- Formation of strata: Difficult problem. Often need subject matter expertise.
  - Some theoretical works, starting from Dalenius (1950).

# Example: Quality control



- A beverage bottling plant producing  $N = 100000$  bottles in a shift.
- Need a sample of  $n = 200$  bottles to test for carbonation levels and seal integrity.
- How do we sample them while the conveyor belt is running?
- Let's try SRS: Serially number the bottles. Choose 200 random numbers.
  - Operationally **difficult**. Need to keep track of the numbers.
  - Hard to grab bottles with close serial numbers without knocking others or stopping the belt.
- Let's try stratified: May stratify by hour of the shift.
  - Similar issues as in SRS.

# Example: Quality control



- Need some **buffer time** between each selected unit.
- One option:
  - Pick one random starting number between 1 and  $\frac{N}{n} = \frac{100000}{200} = 500$ , say, 142.
  - The human/robot arm pushes **bottle #142** off the belt into the testing bin.
  - Then it simply pushes **every 500th bottle** after that. So, #642, #1142, ... and so on, up to #99642. → **sample of size  $n = 200$** .
- No collision issue; robot can count to 500 only and then reset.

**Systematic sampling**

# Systematic sampling: Uses

- When the population is a **flow** rather than a **file**.
  - **Example:** Surveying commuters at the Howrah ferry ghat in a given time range.
- Operationally simpler and faster than SRS. (e.g., no collision issue)
  - In forestry, auditing, or agricultural surveys, it is much easier to instruct a worker to measure **every 10th tree** than to have them navigate to randomly selected GPS coordinates.
- US President John F. Kennedy used to read a systematic sample of letters written to him at the White House.
- Used in 1940 US Population Census to get info on supplementary questions.
- Used in the estimation of catch of marine fish in India. → Central Marine Fisheries Research Institute (CMFRI)
- Used in geological mine surveys, household surveys, record surveys, industrial surveys, spatial surveys.
- Still need a well-defined **sampling frame** to figure out the actual **target population**.
  - **Example:** Sampling every 10th person exiting an airplane.

# Systematic sampling: Setup

- Has **random** component + **non-random** component.
- Let  $N = n \times k$  for some integer  $k \geq 2$ .

## (Linear) Systematic sampling

- Draw a random number from  $\{1, 2, \dots, k\}$ , say  $i$ . That's your first selected unit.
- Select every  $k$ th unit until you get a sample of size  $n$ . ( $k$ : **Sampling interval**)
- **Selected sample**:  $\{i, i + k, i + 2k, \dots, i + (n - 1)k\}$ .

Random start:  $i$

The diagram shows a grid of units arranged in rows and columns. The first row contains units 1, 2, ...,  $i$ , ...,  $k$ . The second row contains units  $k+1$ ,  $k+2$ , ...,  $k+i$ , ...,  $2k$ . The third row contains units  $2k+1$ ,  $2k+2$ , ...,  $2k+i$ , ...,  $3k$ . The fourth row contains units  $\vdots$ ,  $\vdots$ ,  $\vdots$ , ...,  $\vdots$ . The fifth row contains units  $(n-1)k+1$ ,  $(n-1)k+2$ , ...,  $(n-1)k+i$ , ...,  $nk = N$ . A dashed box highlights the column of units  $i, k+i, 2k+i, \dots, (n-1)k+i$ . A blue arrow points down to the unit  $i$  in the first row. A blue bracket at the bottom indicates the systematic sample (every  $k$ -th unit).

|            |            |          |            |     |          |
|------------|------------|----------|------------|-----|----------|
| 1          | 2          | ...      | $i$        | ... | $k$      |
| $k+1$      | $k+2$      | ...      | $k+i$      | ... | $2k$     |
| $2k+1$     | $2k+2$     | ...      | $2k+i$     | ... | $3k$     |
| $\vdots$   | $\vdots$   | $\vdots$ | $\vdots$   |     | $\vdots$ |
| $(n-1)k+1$ | $(n-1)k+2$ | ...      | $(n-1)k+i$ | ... | $nk = N$ |

Systematic sample  
(every  $k$ -th unit)

- Sampling design? 1st and 2nd order inclusion probabilities?

# Remarks

- Systematic sampling is sometimes used as a proxy for SRS.
- If the sampling frame is created in a random order, then a systematic sample **mimics** an SRS.
  - **Example:** List of students is ordered by randomly generating their roll numbers. We want to estimate average student height.
- Does it produce a **representative** sample?
- When the sampling frame is **sorted** in terms of the study variable.
  - Systematic sampling covers the entire spectrum of the population pretty well.
  - **Example:** Selecting every 10th student from an admission list that is sorted from the highest entrance exam score to the lowest.
  - **Example:** Selecting every 20th account from a list of accounts that is sorted from the smallest amount to largest amount.
- When the sampling frame has a **periodic or cyclical** ordering of the study variable.
  - May get a **bad sample**, skewed towards a particular type of observations.
  - **Example:** If males and females alternate in the list, and  $k$  is even, we end up with all males or all females in the sample.
  - **Example:** Estimating average daily road traffic by sampling every 7th day.
- Grouped segregations: Inspecting every 50th tree in a mixed forest where a specific rare disease only spreads in tight groups of 5 or 6 trees.

# Estimation of $\bar{Y}$

## Framework

- $k$ -possible starting points  $\equiv k$ -possible samples.
  - Population divided into  $k$  **Clusters**.
  - Let  $Y_{ij}$  : Value of  $Y$  for unit  $j$  in cluster  $i$ ,  $i \in \{1, 2, \dots, k\}$ ,  $j \in \{1, 2, \dots, n\}$ .
  - Cluster-level mean:  $\bar{Y}_i = \frac{1}{n} \sum_{j=1}^n Y_{ij}$ .
  - Population mean:  $\bar{Y} = \frac{1}{k} \sum_{i=1}^k \bar{Y}_i$ .
- 
- Estimator of  $\bar{Y}$ : sample mean  $\bar{y}$ .
    - $\mathbb{E}(\bar{y}) = \bar{Y}$ . So, sample mean is **unbiased** for  $\bar{Y}$  under systematic sampling.
  - $\text{Var}(\bar{y}) = \frac{1}{k} \sum_{i=1}^k (\bar{Y}_i - \bar{Y})^2 =: \sigma_b^2$  (**between-cluster variance**).
  - Systematic sampling vs SRS: **Which one is more efficient?**
    - Intuitively: If the  $k$  clusters are **similar** to each other, then systematic sampling should work better. (each cluster is a good representative of the population)
    - Clusters similar to each other  $\implies$  **low between-cluster variation**  $\implies$  **high within-cluster variation**.
    - Formally: Introduce the notion of **intraclass** (or intraclass) correlation.

# Intraclass correlation coefficient (ICC)

- Denote  $\rho_c$ : ICC.  $\rightarrow$  A measure of **homogeneity within clusters**.

$$\rho_c = \frac{\frac{1}{kn(n-1)} \sum_{i=1}^k \left[ \sum_{\substack{j=1 \\ j \neq j'}}^n \sum_{j'=1}^n (Y_{ij} - \bar{Y})(Y_{ij'} - \bar{Y}) \right]}{\frac{1}{N} \sum_{i=1}^k \sum_{j=1}^n (Y_{ij} - \bar{Y})^2}.$$

- Within a cluster, stack up all the pairs of observations side-by-side. Do this for all the  $k$  clusters.
  - Here,  $\rho_c$  = Usual Pearson correlation coefficient of the resulting pairs of observations. (not true for unequal sized clusters though)
- Turns out,  $\text{Var}(\bar{y}) = \frac{\sigma^2}{n} \{1 + (n-1)\rho_c\} = \frac{S^2}{n} \left(\frac{N-1}{N}\right) [1 + (n-1)\rho_c]$ .
- Note that  $-\frac{1}{n-1} \leq \rho_c \leq 1$ .
- Compare with SRSWR: Systematic sampling is at least as efficient when  $\rho_c \leq 0$ .
- Compare with SRSWOR: Systematic sampling is at least as efficient when  $\rho_c \leq -\frac{1}{N-1}$ .
- We want the clusters to be **as heterogeneous as possible**, i.e., small  $\rho_c$ . (opposite of what we want in stratified sampling)

# Sample Surveys

Class 9

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# Previously on Sample Surveys...

# Recap

- Systematic sampling.
  - $N = n \times k$ .
  - **Random** start  $\in \{1, 2, \dots, k\}$ . Choose every  $k$ th unit thereafter.
  - $k$ : Sampling interval.
  - Operationally **more convenient** than SRS.
- Systematic sampling: A special case of **Cluster** sampling.
  - The population is divided into  $k$  clusters.
  - Choose a cluster **randomly**.
  - Study **every** unit within the chosen cluster.
- Estimation of  $\bar{Y}$  under systematic sampling.
  - Sample mean  $\bar{y}$  is unbiased.
  - $\text{Var}(\bar{y}) = \sigma_b^2$ : Between-cluster variance.
  - Units within a cluster should be **heterogeneous**.
- Intraclass correlation coefficient  $\rho_c$ .
  - A measure of how units within a cluster **resemble** one another (**internal homogeneity**).
  - $-\frac{1}{n-1} \leq \rho_c \leq 1$ . (so,  $\rho_c$  cannot take the value  $-1$ )
  - $\rho_c \leq 0 \implies$  Systematic sampling is **better** than SRSWR.
  - $\rho_c \leq -\frac{1}{N-1} \implies$  Systematic sampling is **better** than SRSWOR.

# Systematic sampling: Inference for $\bar{Y}$

- $Y_{ij}$  : Value of  $Y$  for unit  $j$  in cluster  $i$ ,  $i \in \{1, 2, \dots, k\}$ ,  $j \in \{1, 2, \dots, n\}$ .
- Cluster-level mean:  $\bar{Y}_i = \frac{1}{n} \sum_{j=1}^n Y_{ij}$ .
- $\text{Var}(\bar{y}) = \frac{1}{k} \sum_{i=1}^k (\bar{Y}_i - \bar{Y})^2 = \sigma_b^2$
- ICC:  $\rho_c = \frac{\frac{1}{kn(n-1)} \sum_{i=1}^k \left[ \sum_{\substack{j=1 \\ j \neq j'}}^n \sum_{j'=1}^n (Y_{ij} - \bar{Y})(Y_{ij'} - \bar{Y}) \right]}{\frac{1}{N} \sum_{i=1}^k \sum_{j=1}^n (Y_{ij} - \bar{Y})^2}$ .
- $\text{Var}(\bar{y}) = \frac{\sigma^2}{n} \{1 + (n-1)\rho_c\} = \frac{S^2}{n} \left(\frac{N-1}{N}\right) [1 + (n-1)\rho_c]$ .

## How to estimate $\text{Var}(\bar{y})$ ?

- We only observe **one** cluster. We don't get to look at other clusters.
  - At the cluster level, we have a sample of **size 1**.
- **Cannot** estimate the variance unbiasedly.
- Available methods: based on some approximations and/or some assumptions.

# Variance estimation

- Recall: Systematic sampling is sometimes used as a proxy for SRS.
  - When the ordering of population units is done **randomly**, then the two sampling designs are equivalent.
- What if we **pretend** that we have an SRS? (say, SRSWOR)

## Estimate the variance as if you have an SRSWOR sample

- $\hat{V} = (1 - f) \frac{s^2}{n}$ , where  $s^2$  is the variance of the sample observations (divisor  $n - 1$ ).
  - Under systematic sampling,  $\mathbb{E}(\hat{V}) = (1 - f) \frac{s^2}{n} \frac{N-1}{N} (1 - \rho_c) \neq \text{Var}(\bar{y})$ .
  - In general, this estimator is **biased**.
    - **Which direction?**
  - Unbiasedness holds iff  $\rho_c = -\frac{1}{N-1}$  (the point where systematic sampling and SRSWOR are equally efficient).
- 
- If you have reasons to believe that the clusters are **heterogeneous enough**, use this method.
    - Get a **conservative** estimator.

# Variance estimation: alternative approaches

- Recall: drawing one cluster (i.e., one systematic sample) isn't enough for estimating between-cluster variation.

## Draw multiple systematic samples from the population.

- Related to the Interpenetrating Network of Subsampling (IPNS) technique of Mahalanobis.
  - Draw  $m(> 1)$  independent systematic samples of size  $n$  from the population. Get the sample mean  $\bar{y}^{(t)}$ ,  $t = 1, 2, \dots, m$ .
  - Unbiased estimator of  $\bar{Y}$ :  $\bar{\bar{y}} = \frac{1}{m} \sum_{t=1}^m \bar{y}^{(t)}$ .
  - Now we can estimate  $\text{Var}(\bar{\bar{y}})$  unbiasedly.
  - Variance estimator:  $\hat{V} = \frac{1}{m(m-1)} \sum_{t=1}^m (\bar{y}^{(t)} - \bar{\bar{y}})^2$ .
- One issue: May not have the budget to collect info on more than  $n$  units.

# Variance estimation: alternative approaches

## Related approach: Use multiple starting points

- Suppose  $n$  is divisible by  $m$ .  $N = \frac{n}{m} \times mk$
  - Choose  $m$  independent random starts from the first  $mk$  units.
  - For each random start, select every  $mk$ -th unit thereafter.
    - So, for each random start, get a systematic sample of size  $\frac{n}{m}$ .
  - We now have  $m$  independent systematic samples of size  $\frac{n}{m}$ . Do the same thing as before.
- 
- Another related approach: Divide the original systematic sample of size  $n$  into  $m$  disjoint subsamples of size  $\frac{n}{m}$  each.
    - Pretend they are **independent systematic samples**. Do the same thing as before.
  - Another approach: assumes an autocorrelation structure on the  $Y_{ij}$ s. (Wolter 1984)
    - $$\hat{V} = \frac{1-f}{n} \left[ \frac{1}{2(n-1)} \sum_{j=1}^{n-1} (Y_{i,j+1} - Y_{ij})^2 \right].$$

# What if $N$ is not a multiple of $n$ ?

## Example: Campus audit

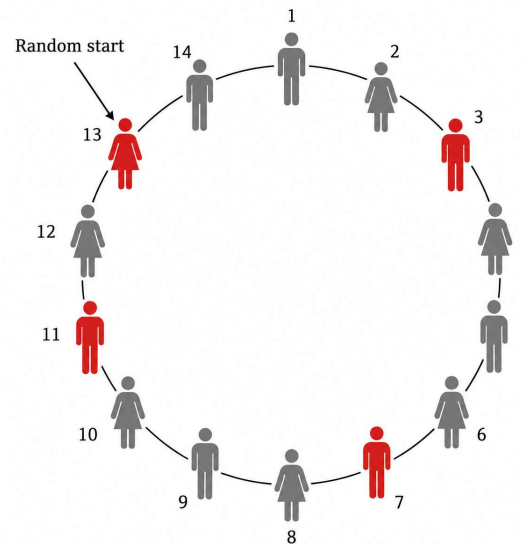
- You are auditing ISI canteen, which has  $N = 14$  staff members.
  - Select a sample of size  $n = 4$  to interview them about food quality, safety protocols, etc.
  - $k = \frac{N}{n} = 3.5 \rightarrow$  **not an integer**.
  - How to do systematic sampling?
- One option: choose  $k = 3$  (could've chosen  $k = 4$  also).



- **No control** over sample size.

# Circular systematic sampling

- Don't arrange the units on a **line**; arrange them on a **circle**.
- Pick a random start from  $\{1, 2, \dots, N\}$ .
- Choose  $k$  to be the nearest integer value of  $\frac{N}{n}$ .
- Here,  $N = 14$ ,  $n = 4$ . Set  $k = 4$ .
- Keep selecting every  $k$ -th unit **along the circle** (go clockwise).



- If random start is  $i$ , then for  $j = 0, 1, 2, \dots, (n - 1)$ ,
  - Select unit  $i + jk$  if  $i + jk \leq N$ .
  - If  $i + jk > N$ , select unit  $i + kj - N$ .